Users' manual for DSurfTomo-4D

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1, Description

DSurfTomo-4D is developed from the original DSurfTomo program for time-lapse surface-wave tomography. It uses two epochs of surface-wave dispersion data and directly inverts the shear-wave velocity change between them.
The original DSurfTomo program is used to obtain the reference velocity model at the first epoch, T1. DSurfTomo-4D then uses the T1 model together with the T1 and T2 dispersion data to recover the velocity perturbation.
For the data format, model format, grid definition, and general input parameters, please refer to the original DSurfTomo user manual.

2, Installation

Download the original DSurfTomo package from:
First, make a directory for the DSurfTomo code and example:
mkdir DSurfTomo
Second, clone the source code from Github with
cd DSurfTomo
git clone https://github.com/HongjianFang/DSurfTomo
At last, install
./configure
After successful compiling, the executable file will be in the 'bin' directory.

Download DSurfTomo-4D from:
git clone https://github.com/ZQ-de/DSurfTomo-4D.git
cd DSurfTomo-4D/src_4D_dev
make
After successful compilation, the executable DSurfTomo-4D will be generated in the src_4D_dev directory.

3, Operation steps

3.1 Synthetic time-lapse test

Synthetic-test mode is used to evaluate the recovery of a prescribed velocity change between two epochs.
The synthetic examples provided in this repository contain all required input files, including the reference model, two-epoch surface-wave dispersion data, and the two true velocity models.

MOD.true_T1 represents the prescribed true velocity model at T1, whereas MOD.true_T2 represents the corresponding true velocity model at T2 after the prescribed velocity change is introduced.

First, use conventional DSurfTomo to invert the synthetic T1 data and obtain DSurfTomo.inMeasure.dat. Then use make_4d_MOD_from_T1.py together with the original MOD template to construct the T1 reference model.
The generated MOD_4D contains the inverted T1 velocity model in the full MOD grid format.
Copy MOD_4D to the 4D working directory and rename it as:
MOD

In DSurfTomo.in, specify the two dispersion-data files, set:
1 c: synthetic flag(0:real data,1:synthetic)
MOD.true_T1
MOD.true_T2
The dispersion-data filenames are user-defined and should match those specified in DSurfTomo.in.

A typical 4D working directory contains:
DSurfTomo.in
MOD
MOD.true_T1
MOD.true_T2
<T1 dispersion data>
<T2 dispersion data>

Run:
./DSurfTomo-4D
The main output is:
DSurfTomo.deltaVs.dat
with four columns:
longitude latitude depth deltaVs

3.2 Real-data inversion
For real-data applications, first use conventional DSurfTomo to invert the T1 dispersion data and obtain:
DSurfTomo.inMeasure.dat
Then use make_4d_MOD_from_T1.py together with the original MOD template to construct the T1 reference model. The generated MOD_4D is copied to the 4D working directory and renamed as MOD.

Prepare the matched common-path dispersion datasets for T1 and T2. The filenames are user-defined and should match the first two lines of DSurfTomo.in.

Set:
0 c: synthetic flag(0:real data,1:synthetic)
No true velocity models are required in real-data mode.
A typical working directory contains:
DSurfTomo.in
MOD
<T1 dispersion data>
<T2 dispersion data>

Run:
./DSurfTomo-4D
The main output is:
DSurfTomo.deltaVs.dat
with four columns:
longitude latitude depth deltaVs
where deltaVs is the recovered shear-wave velocity change between T1 and T2.